WEEK 14 OF 15 · STAGE 3: ADVANCED METHODS

Frontiers and limits

Where IAMs are growing, and where they fall short

New R skill Polishing charts and tables for a report

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	Week 13 Uncertainty	**Week 14 Frontiers and limits**	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Describe three frontiers: air and health, materials, low demand
2. Calculate what happens to warming if planned carbon removal falls short
3. State four common criticisms of IAMs and a fair answer to each
4. Polish project figures: fixed order and colours, finding titles, units

? **Big question** What can IAMs now say about health, materials and energy demand, and which of their habits should make us cautious?

1.5 °C with 40% less energy?

One published scenario meets 1.5 °C with final energy in 2050 about 40% below today, and no technologies that remove CO2 from the air at all.



If that is possible, why do so many pathways rely on removing CO2 later?

KEY IDEA

Three frontiers: IAMs reaching into other fields

1

Air and health

Burning less fossil fuel also cleans the air. Studies set these health gains beside the cost of mitigation (Markandya et al., 2018).

2

Materials

Industrial ecology tracks steel, cement and plastics through the economy. IAMs are starting to link to it (Pauliuk et al., 2017).

3

Low demand

Same services, much less energy: efficient buildings, shared transport, better devices (Grubler et al., 2018).

Worked example: how big is low demand?

Problem The low-demand scenario uses 245 EJ of final energy in 2050, about 40% below today. Roughly how much is used today?

1. 245 EJ is 100% − 40% = 60% of today
2. Today ≈ 245 ÷ 0.6
3. 245 ÷ 0.6 ≈ 408 EJ
4. So the scenario cuts about 408 − 245 = 163 EJ of final energy

Answer

Today is roughly 408 EJ; the scenario uses about 163 EJ less

AN EVERYDAY PICTURE

Carbon removal is like a promise to pay later

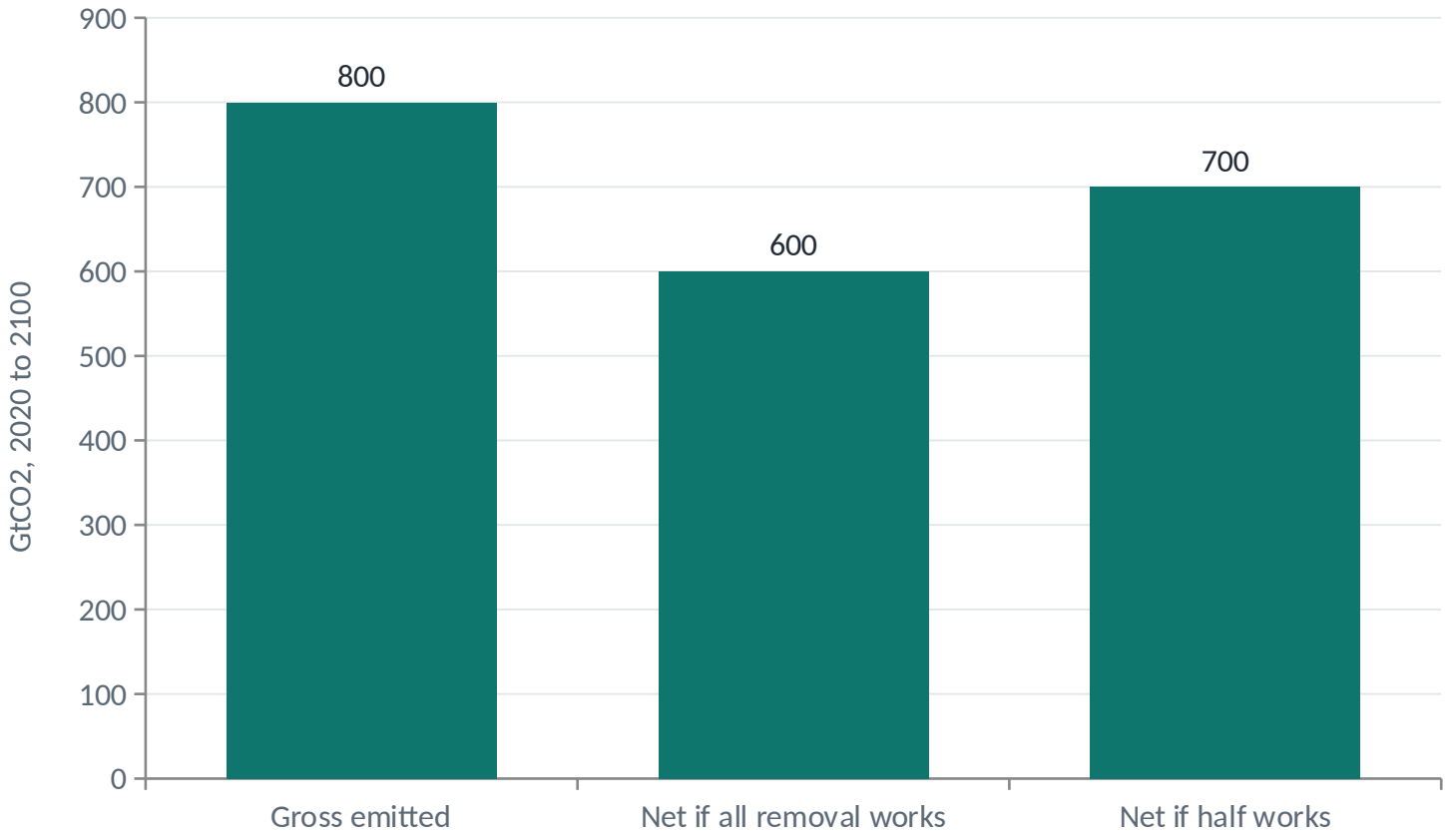
Buying on a credit card lets you have things now and pay later. A pathway that emits now and plans to remove CO2 later works the same way. If the payment never comes, the debt stays: the CO2 is already in the air.

What it means for models

1. Gross emissions: what is put into the air
2. Net emissions: gross minus what is removed
3. Removal not delivered means more warming

PICTURE THE NUMBERS

Undelivered removal leaves more CO2 in the air



Illustrative pathway, not a published scenario

Read it like this

1. Plan: emit 800 Gt, remove 200 Gt
2. Half the removal delivered: 100 Gt more stays in the air
3. The emissions happened; only the clean-up was postponed

Worked example: what if removal falls short?

Problem A pathway emits 800 GtCO2 and plans to remove 200 Gt, but only half the removal is delivered. How much extra warming?

1. Planned net: 800 − 200 = 600 Gt
2. Delivered removal: 200 ÷ 2 = 100 Gt, so net = 800 − 100 = 700 Gt
3. Extra net CO2 = 700 − 600 = 100 Gt
4. Week 2 rule: 0.45 °C per 1,000 Gt
5. Extra warming = 0.45 × 100 ÷ 1,000 = 0.045 °C

Answer **About 0.045 °C extra; a 400 Gt shortfall would add about 0.18 °C**

SIDE BY SIDE

Four criticisms every IAM user should know

Criticism	The point	Source
Reliance on carbon removal	If removal falls short, warming is higher	Anderson and Peters (2016)
Equity	Cheapest cuts, but who pays is left open	Week 4: discounting
Transparency	Hard to inspect; open code helps	Gambhir et al. (2019)
Assumptions that age	Models kept underestimating solar growth	Creutzig et al. (2017)

Give each criticism its strongest form before answering it

Quick check

A pathway plans 300 Gt of removal but only a third is delivered. Extra warming at 0.45 °C per 1,000 Gt?

- **A** 0.045 °C
- **B** 0.09 °C
- **C** 0.135 °C
- **D** 0.3 °C

Quick check: answer

A pathway plans 300 Gt of removal but only a third is delivered. Extra warming at 0.45 °C per 1,000 Gt?

A. 0.045 °C

B. **0.09 °C** ✓

C. 0.135 °C

D. 0.3 °C

HOW IT WORKS

Four questions to ask of any IAM result

1

What kind of model?

Simulation or optimisation, foresight, energy only or whole economy

2

What was assumed?

Technology costs, energy demand, how much carbon removal is allowed

3

One run or many?

Look for sensitivity tests, an ensemble, a range across models

4

What is left out?

Regions, future generations, effects the model does not count

Builds on week 6's questions for reading a scenario chart. Apply it to your own project first.

COMPARE

Project clinic: a chart that says something

Before

- Title names the variable: 'CO2 by scenario'
- Scenarios in alphabetical order, colours change
- No units on the axis; size depends on the laptop

After

- Title states the finding: 'Net zero halves 2040 emissions'
- Reference first, same colour in every figure
- Units on the axis; saved at 16 × 10 cm, 300 dpi

WATCH OUT

Common mix-ups

Not quite	Better
The criticisms mean IAMs are useless	They tell you which questions to ask. One review's verdict: keep the models and supplement them.
Removal later cancels emissions now, so the timing does not matter	The CO2 is in the air until removal is delivered, and delivery is uncertain.
A chart title should name the variable	It should state the finding, with the variable and units on the axis.

NEW WORDS THIS WEEK

Words to add to the class glossary

Co-benefit A gain that comes with climate action without being its goal, such as cleaner air from burning less coal.

Industrial ecology The study of how materials such as steel and cement flow through the economy, using tools like life cycle assessment.

Low-demand scenario A pathway where people get the same services with much less energy, through efficiency and changes in how we live and travel.

Carbon removal Taking CO2 back out of the air and storing it, for example with forests or bioenergy plus carbon capture.

Equity Fairness: who pays for cutting emissions, who suffers the damages, and how present and future generations are weighed.

Lab 14: project clinic

Bring your draft results and polish them: one scenario order and colour set, titles that state findings, units, figures saved at a fixed size and a clean summary table.

Script: labs/lab14.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Fix one order and one colour set

Define these once at the top of the script, or in R/style.R, and use them in every figure.

```
scen_order <- c("Reference", "CheapSolar",
                "CarbonPrice", "NetZero")
scen_cols <- c(Reference   = "#5E6E73",
               CheapSolar  = "#E69F00",
               CarbonPrice = "#0072B2",
               NetZero     = "#009E73")
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

Put the scenarios in order

Turn scenario into a factor with your order. Fill GAP 1 with the order vector.

```
co2 <- getQuery(prj, "CO2 emissions by region") |>
  filter(region == "Japan", year >= 2020) |>
  mutate(mt = value * 44 / 12)   # MTC to MtCO2
co2 <- co2 |>
  mutate(scenario = factor(scenario,
                           levels = ___))  # GAP 1
```

Fill each ___ gap before running that line. Stuck? The solution file has the complete version.

Write a title that states the finding

Fill GAP 2 with your finding in one line. Units go on the axis, the source in the caption.

```
p <- ggplot(co2, aes(year, mt, colour = scenario)) +
  geom_line(linewidth = 1) +
  scale_colour_manual(values = scen_cols) +
  labs(title = "___",                  # GAP 2
       x = NULL,
       y = "CO2 emissions (MtCO2 per year)",
       colour = NULL,
       caption = "Source: own GCAM runs.") +
  theme_minimal(base_size = 14)
```

Save at a fixed size and make a table

Fix width, height and resolution in code. Fill GAP 3 with 300 for print quality.

```
ggsave("output/fig1_co2.png", plot = p,
       width = 16, height = 10,
       units = "cm", dpi = ___)   # GAP 3
tab <- co2 |>
  filter(year %in% c(2030, 2050)) |>
  select(scenario, year, mt) |>
  pivot_wider(names_from = year, values_from = mt)
knitr::kable(tab, digits = 0,
  caption = "CO2 emissions, MtCO2 per year")
```

Did it work?

You should see

- The legend lists scenarios in your order, Reference first
- Each figure title states a finding, and the axis shows units
- output/fig1_co2.png exists and the table has no decimals

Worksheet questions

1. Write the title of your project's main figure as a finding, and say which number in your results supports it.
2. Apply the four-question checklist to your own project in four sentences.
3. Which of the four criticisms applies most to your study, and how will your report say so?

RECAP

Take-home messages

1. IAMs are reaching into air quality and health, materials, and low-demand futures
2. Removal not delivered leaves CO2 in the air: 0.45 °C per 1,000 Gt shortfall
3. Four criticisms: reliance on removal, equity, transparency and assumptions that age
4. Read every result, including your own, with the four-question checklist
5. Good project figures state a finding, keep one order and colour set, and show units

Next: week 15

Week 15 is presentation day: each team presents its scenario study, and we review the whole course.

Practise

Interactive course, week 14: try the cdr explorer and the 5-question quiz.

Readings: Pauliuk et al. (2017), Industrial ecology in integrated assessment models, Nature Climate Change; Anderson and Peters (2016), The trouble with negative emissions, Science; Optional: Gambhir et al. (2019), Energies